

Technical Document #373: Blockchain - Comprehensive Overview

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Proof of Work (PoW) is a consensus mechanism where miners solve complex puzzles to validate transactions. It is energy-intensive but secure.

Tokenomics studies the economic systems of cryptocurrency tokens. It covers supply mechanisms, incentive structures, and value capture.

Consensus mechanisms ensure all nodes in a network agree on the state of the blockchain. They are essential for maintaining network integrity.

Decentralized finance (DeFi) recreates traditional financial systems using blockchain technology. It enables lending, borrowing, and trading without intermediaries.

Augmented reality overlays digital information on the physical world. It has applications in training, maintenance, and entertainment.

Natural language processing has made significant advances with large language models. These models can understand and generate human-like text with remarkable fluency.

Cloud computing has democratized access to powerful computing resources. Startups can now access the same infrastructure as large enterprises.

Edge computing processes data closer to its source, reducing latency and bandwidth usage. It is essential for real-time applications like autonomous vehicles.

Autonomous vehicles use a combination of sensors, AI, and mapping to navigate without human intervention. Self-driving technology is rapidly advancing.

Key Takeaways:

- Non-fungible tokens (NFTs) represent unique digital assets on the blockchain.
- Tokenomics studies the economic systems of cryptocurrency tokens.
- Proof of Work (PoW) is a consensus mechanism where miners solve complex puzzles to validate transactions.